- PDE ELLICHE
  - COSTO COMPUTAZIONALE CRESCENTI
    - MESH FINI
    - DOMINI COMPLESSI
    - SOLUZIONE MOLTO ACCURATA

    # DOF (DEGREES OF FREEDOM)

    SOLUZIONE SYS LINEARE $\mathcal{O}(\#DOF^2)$

    $\mathcal{O}(\#DOF^3)$

- RIDURRE IL COSTO COMPUTAZIONALE

PDE PARAMETRICA

$$a(u_h, v_h; \mu) = F(v_h; \mu) \quad \forall v_h \in V_h(\mu)$$

$$+ BC(\mu)$$

$$\mu \in \mathbb{P} \subset \mathbb{R}^p$$

- OTTIMIZZAZIONE TANTE SOLUZIONI PER PARAMETRI $\neq$

- FASE OFFLINE (TANTO TEMPO)
  QUALCHE SOLUZIONE PER QUALCHE PARAMETRO

- FASE ONLINE (VELOCI) DATO UN NUOVO PARAMETRO VELOCI UNA SOLUZIONE

  SPAZIO RIDOTTO

- CACCIA $V_n = \langle \{\psi_i\}_{i=1}^{n} \rangle$ $\quad n \ll N_h$

ELEMENTI FINITI

$$V_h = \langle \{\varphi_i\}_{i=1}^{N_h} \rangle$$

$V_m \subseteq V_h \qquad u_h(\mu) \in V_m \qquad \mu \in \mathbb{P}$

3. COME TROVIAMO V_m?

2. IL PROBLEMA IN V_m È PIÙ VELOCE? (INDIPENDENTE DA N_h)

1. QUALE È IL PROBLEMA RIDOTTO?

<u>FEM</u>

$$u_h(x) = \sum_{j=1}^{N_h} u_j \, \varphi_j(x)$$

$$\sum_j \underbrace{a(\varphi_i, \varphi_j; \mu)}_{\underline{\underline{A}}(\mu)} \cdot u_j = F(\varphi_i; \mu) \qquad \forall \varphi_i \quad \forall i = 1, \dots, N_h$$

$$\underline{\underline{A}}(\mu) \cdot \underline{u} = \underline{f}(\mu)$$

<u>ROM</u>

$$V_m = \langle \{\psi_i\}_{i=1}^{m} \rangle$$

$$\underline{u_m}(x) := \sum_{j=1}^{m} \psi_j(x) \cdot \hat{u}_j \qquad A \in \mathbb{R}^{N_h \times N_h}$$

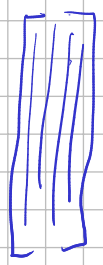
$$\sum_j a(\psi_i, \psi_j; \mu) \cdot \hat{u}_j = F(\psi_i; \mu)$$

$$\hat{\underline{\underline{A}}}(\mu) \, \hat{\underline{u}} = \hat{\underline{f}}(\mu)$$

$$\hat{A} \in \mathbb{R}^{m \times m}$$

$\in V_m \subseteq V_h$

$$\psi_i(x) = \sum_{j=1}^{N_h} B_{ji} \, \varphi_j(x) \qquad B \in \mathbb{R}^{N_h \times m}$$



$B_{:i}$ = VETTORE DEI COEFFICIENTI DELLA FUNZIONE DI BASE

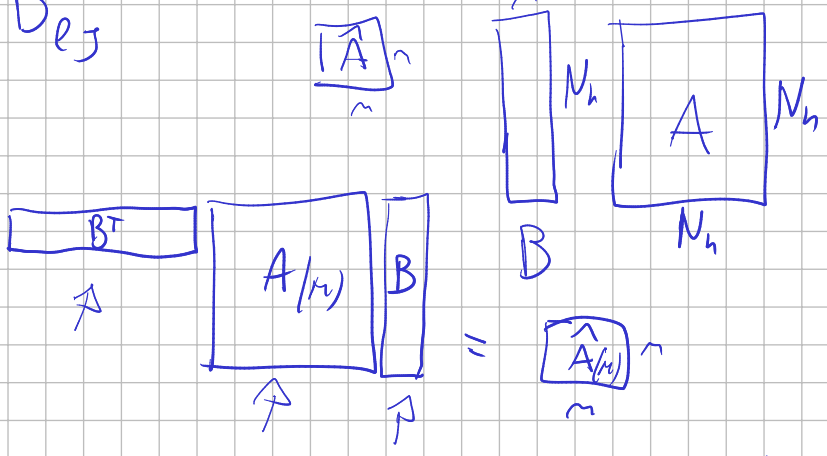
$$\hat{A}(\mu)_{ij} = a(\psi_i, \psi_j; \mu)$$

DELLO SPAZIO RIDOTTO i-esima NELLO SPAZIO DEGLI ELEMENTI FINITI

$$= a\left(\sum_{k=1}^{N_h} B_{ki}\, \varphi_k, \sum_{l=1}^{N_h} B_{lj}\, \varphi_l; \mu\right)$$

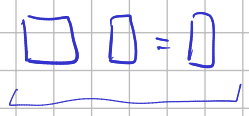
$$= \sum_{\substack{k=1\\l=1}}^{N_h} B_{ki}\, \underbrace{a(\varphi_k, \varphi_l; \mu)}_{A_{kl}}\, B_{lj}$$

$$= B^T A B =$$



$$\hat{f}_i(\mu) = F(\psi_i; \mu) = \sum_{l=1}^{N_h} B_{li}\, F(\varphi_l; \mu) = B^T \cdot \underline{f} = \hat{\underline{f}}$$

$$\hat{A}\, \hat{u} = \hat{F}$$



DIMENSIONE n ! PERFETTO !

RISOLUZIONE DEL SISTEMA ✓ RIDOTTA

- <u>IPOTESI</u> IL PROBLEMA è AFFINE RISPETTO AL PARAMETRO μ

$$A(\mu) = \sum_{q=1}^{Q_a} \underbrace{\Theta_a^q(\mu)}_{\in \mathbb{R}} \cdot \underbrace{A_q}_{\in \mathbb{R}^{N_h \times N_h}} \qquad F(\mu) = \sum_{q=1}^{Q_f} \underbrace{\Theta_q^F(\mu)}_{\in \mathbb{R}} \cdot \underbrace{F_q}_{\in \mathbb{R}^{N_h}}$$

$$\Theta_a^q : \mathbb{P} \to \mathbb{R}$$

$$\hat{A}(\mu) = B^T A(\mu) B = B^T \sum_{q=1}^{Q_a} \Theta_q^a(\mu) A_q B$$

$$= \sum_{q=1}^{Q_a} \Theta_q^a(\mu) \underbrace{(B^T A_q B)}_{\hat{A}_q}$$

$$= \sum_{q=1}^{Q_a} \Theta_q^a(\mu) \hat{A}_q$$

- FASE OFFLINE (ABBIAMO TEMPO) ASSEMBLIAMO A_q ←

  RIDUCIAMO $\hat{A}_q \in \mathbb{R}^{n \times n}$

  $\hat{A}$

- FASE ONLINE (POCO TEMPO)

  SOMMIAMO $\sum_{q=1}^{Q_a} \Theta_q^a(\mu) \hat{A}_q \in \mathbb{R}^{n \times n}$ ←

  VELOCE ✓

ESEMPIO

$$a(\varphi_i, \varphi_j; \mu) = \int_\Omega K(x;\mu) \cdot \nabla\varphi_j \cdot \nabla\varphi_i \, dx$$

$$= \mu_1 \underbrace{\int_{\Omega_1} \nabla\varphi_j \nabla\varphi_i \, dx}_{} + 1 \cdot \int_{\Omega_2} \nabla\varphi_j \cdot \nabla\varphi_i \, dx$$

$$= \Theta_1^a(\mu_1) (A_1)_{ij} + \underset{1}{\underset{\shortparallel}{\Theta_2^a(\mu_1)}} \cdot (A_2)_{ij}$$

$$f(\varphi_i) = \int_{\Gamma_{BASE}} \mu_2 \varphi_i \, ds = \underset{\Theta_1^f(\mu)}{\mu_2} \cdot \underset{f_1}{\int_{\Gamma_{BASE}} \varphi_i \, ds}$$

TROVARE V_n

PROPER ORTHOGONAL DECOMPOSITION (POD)

$\{\mu_i\}_{i=1}^{N_s}$ $\qquad S = [u_h(\mu_1), \dots u_h(\mu_{N_s})] \in \mathbb{R}^{N_h \times N_s}$

COSTOSA (FASE OFFLINE)

$N_s \ll N_h$

<u>SVD</u>

$$S = U \Sigma V^T$$

$U \in \mathbb{R}^{N_h \times N_s}$ $\qquad \Sigma \in \mathbb{R}^{N_s \times N_s}$ $\qquad V^T \in \mathbb{R}^{N_s \times N_s}$

$$\Sigma = \begin{bmatrix} \sigma_1 & & & 0 \\ & \sigma_2 & & \\ & & \ddots & \\ 0 & & & \sigma_{N_s} \end{bmatrix}$$

$\sigma_1 \geq \sigma_2 \geq \dots \geq \sigma_{N_s} \geq 0$

$m \leq N_s$

$$\hat{\Sigma} = \begin{bmatrix} \sigma_1 & & \\ & \ddots & \\ & & \sigma_m \end{bmatrix}$$

$$\hat{V} = [v_1 \dots v_m]$$

$$V = [v_1 \dots v_m \dots v_{N_s}]$$

$$U = [\varphi_1 \dots \varphi_m \dots \varphi_{N_s}]$$

$$\hat{U} = [\varphi_1 \dots \varphi_m] \in \mathbb{R}^{N_h \times m}$$

$$\boxed{N_s}$$

$$\underbrace{S}_{} \approx \underbrace{\hat{S}}_{} = \underbrace{\hat{U}}_{\in \mathbb{R}^{N_h \times m}} \hat{\Sigma} \hat{V}^T$$

$$\sum_{i=1}^{m} \underbrace{\psi_i}_{\in \mathbb{R}^{N_h}} \cdot s_i$$

$$B = \hat{U}$$

$$\psi_i = \hat{U}_{:,i}$$

$$V_m = \langle \{\psi_i\}_{i=1}^{m} \rangle$$

COSTO COMPUTAZIONALE DI SVD $\quad S \in \mathbb{R}^{N_h \times N_s}$

$$N_s \ll N_h$$

- SETTIMANA PROX PARABOLICI = ELLITTICI + TEMPO
- MAGGIO IPERBOLICI